

KALAM CX

Data Schema & Entity Relationship Design (ERD)

AI-Powered After-Call Automation Platform

Project Name	AI-Powered After-Call Automation Platform
Document Title	Data Schema & Entity Relationship Design (ERD)
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1. Database Overview

Purpose

The data schema exists to support a reliable, auditable, and non-hallucinating record of every customer support call handled through the AI-Powered After-Call Automation Platform. It provides the structural foundation for customer lookup, documentation retrieval, AI-based extraction, classification, business rule validation, human review, final storage, and operational analytics.

Scope

The schema covers the full lifecycle of a single call, from the moment a completed transcript enters the system through final approved storage and its inclusion in aggregated analytics. It spans customer and agent identity, call metadata, AI-generated documentation, ticketing, human review, reference/lookup data, and reporting.

Goals

- Guarantee that every stored fact can be traced back to either the transcript, a human reviewer, or an explicit business rule.
- Prevent fabricated or inferred data by using the explicit value Not Stated wherever information is unavailable.
- Provide a Human-in-the-Loop control point for every generated record prior to final approval.
- Support multi-tenant operation across many client companies such as Kalam CX and its customers.
- Enable operational analytics without degrading transactional performance.

Why the Schema Is Normalized

The schema is organized in Third Normal Form (3NF) so that each fact is stored exactly once. Reference data such as Categories, Sentiments, and Dispositions is separated from transactional data such as CallRecords, eliminating update anomalies and ensuring classification values remain consistent across the platform.

Scalability, Maintainability & Auditing

Normalization keeps transactional tables narrow while a dedicated OperationalStatistics table absorbs reporting cost. Surrogate keys, targeted indexing, and tenant partitioning through ClientCompanies allow horizontal scale. Clear entity boundaries mean changes to one area, such as adding a disposition, need no structural changes elsewhere. Every CallRecords row carries created_at/updated_at timestamps, and every review action is logged in ReviewQueue with reviewer identity and edited fields — together preserving a complete, reconstructable history.

2. Database Design Principles

Normalization

Entities are decomposed so non-key attributes depend only on their own table's primary key, eliminating redundant storage of customer, agent, and reference data.

Referential Integrity

Every relationship is enforced through foreign key constraints, so a CallRecords row can never reference a non-existent Call, Customer, Category, or Reviewer.

Data Consistency

Controlled vocabularies (Categories, Subcategories, Sentiments, Dispositions) are centralized in lookup tables so identical concepts are represented identically platform-wide.

Auditability

Timestamps, reviewer identity, and edited-field tracking are first-class attributes rather than an afterthought (see Section 15).

Scalability

Surrogate keys, selective indexing, and a separated analytics table let the schema support millions of calls without redesign (see Section 17).

Security

Sensitive data is isolated by client company and governed through Role-Based Access Control (see Section 16).

Performance

Indexes are aligned to primary access patterns: customer lookup, review queue processing, ticket search, and category filtering (see Section 12).

Future Extensibility

New reference values, review levels, or AI model outputs can be added by extending lookup tables or adding nullable columns, without breaking existing data.

3. High-Level Data Flow

The platform processes each completed call through a fixed sequence of stages. The schema is designed so every stage writes to, or reads from, a clearly defined set of tables, keeping the flow traceable end to end.

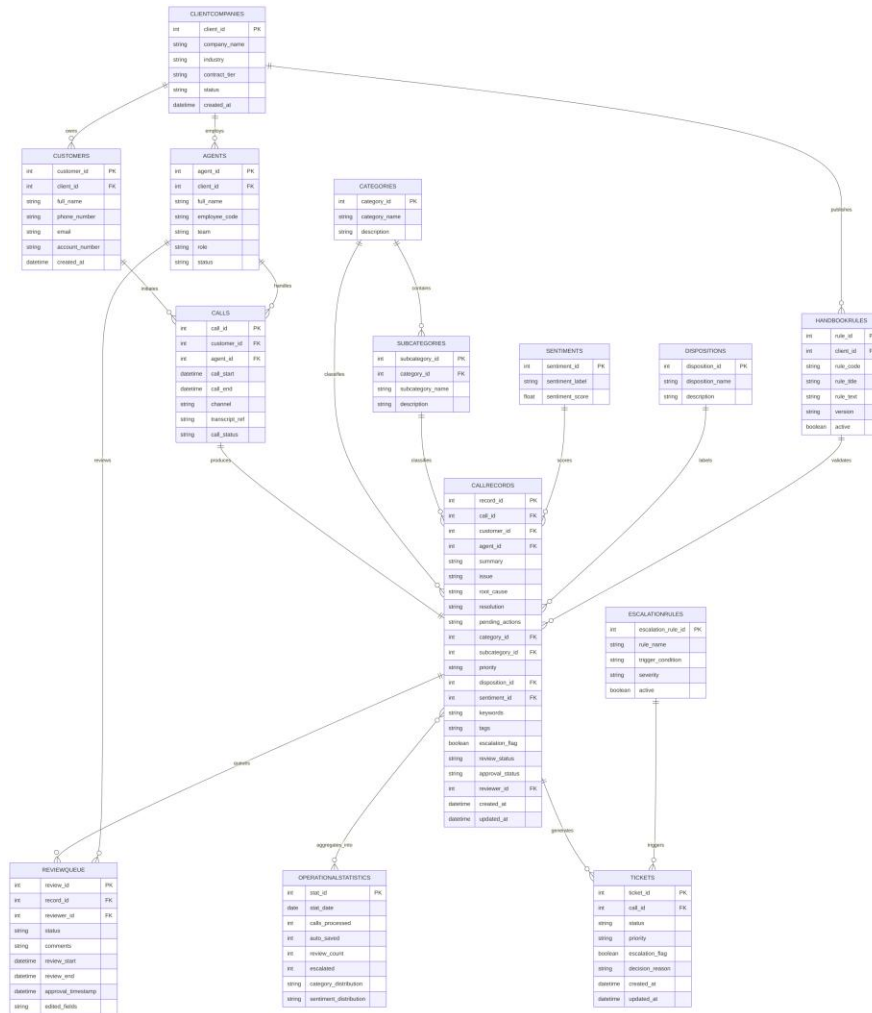
- 1. Transcript Ingestion
- 2. Customer Lookup
- 3. Documentation Retrieval
- 4. AI Extraction
- 5. Classification
- 6. Business Rule Validation
- 7. Human Review
- 8. Final Record Storage
- 9. Operational Analytics

Stage-to-Table Mapping

Stage	Primary Tables Involved
Transcript Ingestion	Calls
Customer Lookup	Customers, ClientCompanies
Documentation Retrieval	HandbookRules
AI Extraction	CallRecords
Classification	Categories, Subcategories, Sentiments, Dispositions
Business Rule Validation	EscalationRules, Tickets
Human Review	ReviewQueue, Agents
Final Record Storage	CallRecords
Operational Analytics	OperationalStatistics

4. Entity Relationship Diagram

The diagram below shows all fourteen entities in the schema and their relationships.



5. Entity Descriptions

This section documents every entity in the schema: purpose, columns, keys, constraints, indexes, relationships, business rules, and a representative example record.

5.1 ClientCompanies

Stores the organizations (contact centers or brands) that subscribe to the After-Call Automation Platform.

Top-level tenant entity. Every Customer, Agent, and HandbookRule is scoped to a client company, enabling isolated multi-tenant operation.

Column	Data Type	Null	Default	Key
client_id	INT (Identity)	No	Auto-generated	PK
company_name	VARCHAR(200)	No	None	-
industry	VARCHAR(100)	Yes	Not Stated	-
contract_tier	VARCHAR(50)	No	Standard	-
status	VARCHAR(20)	No	Active	-
created_at	DATETIME	No	Current timestamp	-

Primary Key	client_id
Foreign Keys	None
Unique Constraints	company_name (per active tenant)
Indexes	IX_ClientCompanies_Status on status
Relationships	1:N to Customers, Agents, and HandbookRules.
Business Rules	A client company must be Active before new calls can be processed. Deactivation blocks new intake but never deletes historical data.

Example: client_id: 100 | company_name: Kalam CX Retail Division | industry: Retail Banking | contract_tier: Enterprise | status: Active

5.2 Customers

Stores identifying and contact information for the end customers who place support calls.

Master record used during Customer Lookup, the first automated step of the after-call workflow.

Column	Data Type	Null	Default	Key
customer_id	INT (Identity)	No	Auto-generated	PK
client_id	INT	No	None	FK
full_name	VARCHAR(150)	No	None	-
phone_number	VARCHAR(30)	Yes	Not Stated	-
email	VARCHAR(150)	Yes	Not Stated	-
account_number	VARCHAR(50)	Yes	Not Stated	-
created_at	DATETIME	No	Current timestamp	-

Primary Key	customer_id
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Foreign Keys	client_id references ClientCompanies(client_id)
Unique Constraints	(client_id, account_number), where account_number is not Not Stated
Indexes	IX_Customers_Phone, IX_Customers_Account, IX_Customers_Client
Relationships	N:1 to ClientCompanies; 1:N to Calls.
Business Rules	Unmatched lookups store Not Stated rather than an assumed identity. Identity fields are never overwritten by AI extraction — only verified profile updates.

Example: customer_id: 58211 | client_id: 100 | full_name: Sara Ahmed | phone_number: +20-100-555-0142 | account_number: ACC-88421

5.3 Agents

Stores the support agents who handle calls and, where applicable, act as reviewers.

Represents both front-line staff and the reviewer pool used in the Human-in-the-Loop workflow.

Column	Data Type	Null	Default	Key
agent_id	INT (Identity)	No	Auto-generated	PK
client_id	INT	No	None	FK
full_name	VARCHAR(150)	No	None	-
employee_code	VARCHAR(30)	No	None	-
team	VARCHAR(100)	Yes	Not Stated	-
role	VARCHAR(50)	No	Agent	-
status	VARCHAR(20)	No	Active	-

Primary Key	agent_id
Foreign Keys	client_id references ClientCompanies(client_id)
Unique Constraints	(client_id, employee_code)
Indexes	IX_Agents_EmployeeCode, IX_Agents_Role
Relationships	N:1 to ClientCompanies; 1:N to Calls (as handler); 1:N to ReviewQueue (as reviewer).
Business Rules	Only Reviewer or Supervisor roles may be assigned in ReviewQueue.reviewer_id. Inactive agents cannot take new calls or reviews.

Example: agent_id: 340 | client_id: 100 | full_name: Omar Nabil | employee_code: EMP-2245 | role: Reviewer | status: Active

5.4 Calls

Stores metadata for each completed customer support call, including the transcript reference.

The intake entity that anchors the after-call workflow; parent of exactly one CallRecords row.

Column	Data Type	Null	Default	Key
call_id	INT (Identity)	No	Auto-generated	PK
customer_id	INT	Yes	Not Stated	FK

Column	Data Type	Null	Default	Key
agent_id	INT	No	None	FK
call_start	DATETIME	No	None	-
call_end	DATETIME	No	None	-
channel	VARCHAR(30)	No	Voice	-
transcript_ref	VARCHAR(300)	No	None	-
call_status	VARCHAR(30)	No	Completed	-

Primary Key	call_id
Foreign Keys	customer_id → Customers(customer_id); agent_id → Agents(agent_id)
Unique Constraints	transcript_ref
Indexes	IX_Calls_CustomerId, IX_Calls_AgentId, IX_Calls_CallStart
Relationships	N:1 to Customers (nullable) and Agents; 1:1 to CallRecords.
Business Rules	Only Completed calls are eligible for AI extraction. transcript_ref must point to an immutable, archived transcript.

Example: call_id: 900321 | customer_id: 58211 | agent_id: 340 | channel: Voice | call_status: Completed

5.5 CallRecords

Stores the structured, AI-extracted, and human-reviewed documentation produced for each call — the core business entity of the platform.

Holds every field produced by AI extraction and classification, plus the review/approval state required by the Human-in-the-Loop workflow. See Section 9 for full field list.

Column	Data Type	Null	Default	Key
record_id	INT (Identity)	No	Auto-generated	PK
call_id	INT	No	None	FK
customer_id	INT	Yes	Not Stated	FK
agent_id	INT	No	None	FK
category_id	INT	Yes	Not Stated	FK
subcategory_id	INT	Yes	Not Stated	FK
disposition_id	INT	Yes	Not Stated	FK
sentiment_id	INT	Yes	Not Stated	FK
reviewer_id	INT	Yes	Not Stated	FK

Primary Key	record_id
Foreign Keys	call_id, customer_id, agent_id, category_id, subcategory_id, disposition_id, sentiment_id, reviewer_id — each references its parent table
Unique Constraints	call_id (one CallRecord per Call)
Indexes	IX_CallRecords_CallId, IX_CallRecords_ReviewStatus, IX_CallRecords_Category, IX_CallRecords_CreatedAt

Relationships	1:1 to Calls; N:1 to Categories, Subcategories, Sentiments, Dispositions; 1:N to Tickets and ReviewQueue.
Business Rules	No field may be inferred or fabricated — unsupported values store Not Stated. approval_status cannot become Approved until review_status is Reviewed. escalation_flag is set automatically whenever EscalationRules criteria are met.

Example: record_id: 771001 | call_id: 900321 | issue: Card decline | resolution: Not Stated | review_status: Pending | approval_status: Pending

5.6 Tickets

Stores the follow-up work item created when a call requires action beyond the call itself, including the Decision Engine's reasoning.

Records the outcome of Business Rule Validation — status, priority, and a decision_reason explaining the routing decision.

Column	Data Type	Null	Default	Key
ticket_id	INT (Identity)	No	Auto-generated	PK
call_id	INT	No	None	FK
status	VARCHAR(30)	No	Open	-
priority	VARCHAR(20)	No	Normal	-
escalation_flag	BOOLEAN	No	False	-
decision_reason	VARCHAR(300)	No	None	-
created_at	DATETIME	No	Current timestamp	-
updated_at	DATETIME	No	Current timestamp	-

Primary Key	ticket_id
Foreign Keys	call_id references Calls(call_id)
Unique Constraints	None
Indexes	IX_Tickets_Status, IX_Tickets_CallId, IX_Tickets_Priority
Relationships	N:1 to Calls/CallRecords; N:1 (optional) to EscalationRules.
Business Rules	decision_reason must always state a human-readable explanation (e.g. High Priority, Negative Sentiment, Incomplete Information, Policy Sensitive Case, Customer History). A ticket cannot close while its CallRecords approval_status is Pending.

Example: ticket_id: 55010 | call_id: 900321 | status: Open | priority: High | decision_reason: Negative Sentiment

5.7 ReviewQueue

Stores every human review task, tracking who reviewed a record, what was changed, and when it was approved.

Implements the mandatory Human-in-the-Loop control point. No CallRecords row may reach final approval without a matching Approved entry here.

Column	Data Type	Null	Default	Key
review_id	INT (Identity)	No	Auto-generated	PK
record_id	INT	No	None	FK
reviewer_id	INT	No	None	FK

Column	Data Type	Null	Default	Key
status	VARCHAR(30)	No	Pending	-
comments	VARCHAR(1000)	Yes	Not Stated	-
review_start	DATETIME	Yes	Not Stated	-
review_end	DATETIME	Yes	Not Stated	-
approval_timestamp	DATETIME	Yes	Not Stated	-
edited_fields	VARCHAR(500)	Yes	Not Stated	-

Primary Key	review_id
Foreign Keys	record_id → CallRecords(record_id); reviewer_id → Agents(agent_id)
Unique Constraints	None
Indexes	IX_ReviewQueue_Status, IX_ReviewQueue_RecordId, IX_ReviewQueue_Reviewer
Relationships	N:1 to CallRecords; N:1 to Agents (as reviewer).
Business Rules	edited_fields must list every field changed from the AI draft, preserving the audit trail. approval_timestamp is populated only once status becomes Approved.

Example: review_id: 40007 | record_id: 771001 | reviewer_id: 340 | status: Approved | edited_fields: resolution, tags

5.8 HandbookRules

Stores the documentation, policy, and handbook content used for retrieval during the Documentation Retrieval step.

The knowledge base referenced when validating AI-extracted information against client-specific policy. Rules are versioned for auditability.

Column	Data Type	Null	Default	Key
rule_id	INT (Identity)	No	Auto-generated	PK
client_id	INT	No	None	FK
rule_code	VARCHAR(50)	No	None	-
rule_title	VARCHAR(200)	No	None	-
rule_text	VARCHAR(MAX)	No	None	-
version	VARCHAR(20)	No	1.0	-
active	BOOLEAN	No	True	-

Primary Key	rule_id
Foreign Keys	client_id references ClientCompanies(client_id)
Unique Constraints	(client_id, rule_code, version)
Indexes	IX_HandbookRules_Code, IX_HandbookRules_Active
Relationships	N:1 to ClientCompanies; logically N:N to CallRecords (resolved through validation, not a junction table).

Business Rules	Only active rules are used to validate new calls; deactivated versions remain for audit. Changes to rule_text create a new version rather than overwriting history.
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Example: rule_id: 12 | client_id: 100 | rule_code: POL-REFUND-01 | rule_title: Refund Eligibility Policy | version: 2.1 | active: True

5.9 Categories

Stores the top-level classification values used to categorize a call's subject matter.

The controlled vocabulary used by the AI Classification stage; parent of Subcategories and referenced directly by CallRecords.

Column	Data Type	Null	Default	Key
category_id	INT (Identity)	No	Auto-generated	PK
category_name	VARCHAR(100)	No	None	-
description	VARCHAR(300)	Yes	Not Stated	-

Primary Key	category_id
Foreign Keys	None
Unique Constraints	category_name
Indexes	IX_Categories_Name
Relationships	1:N to Subcategories; 1:N to CallRecords.
Business Rules	A category cannot be deleted while active Subcategories or CallRecords reference it.

Example: category_id: 4 | category_name: Billing Dispute | description: Charges questioned by the customer

5.10 Subcategories

Stores the second-level classification values that refine a Categories value.

Enables finer-grained classification and reporting for category filtering.

Column	Data Type	Null	Default	Key
subcategory_id	INT (Identity)	No	Auto-generated	PK
category_id	INT	No	None	FK
subcategory_name	VARCHAR(100)	No	None	-
description	VARCHAR(300)	Yes	Not Stated	-

Primary Key	subcategory_id
Foreign Keys	category_id references Categories(category_id)
Unique Constraints	(category_id, subcategory_name)
Indexes	IX_Subcategories_CategoryId
Relationships	N:1 to Categories; 1:N to CallRecords.
Business Rules	A subcategory must always belong to a valid, active category.

Example: subcategory_id: 21 | category_id: 4 | subcategory_name: Duplicate Charge

5.11 Sentiments

Stores the controlled set of sentiment labels applied to a call by the AI extraction pipeline.

A lookup of sentiment classifications (e.g. Positive, Neutral, Negative) with a numeric score, used for both classification and reporting.

Column	Data Type	Null	Default	Key
sentiment_id	INT (Identity)	No	Auto-generated	PK
sentiment_label	VARCHAR(30)	No	None	-
sentiment_score	DECIMAL(4,2)	Yes	Not Stated	-

Primary Key	sentiment_id
Foreign Keys	None
Unique Constraints	sentiment_label
Indexes	IX_Sentiments_Label
Relationships	1:N to CallRecords.
Business Rules	sentiment_score, when present, must fall between -1.00 and 1.00.

Example: sentiment_id: 3 | sentiment_label: Negative | sentiment_score: -0.62

5.12 Dispositions

Stores the controlled set of outcome labels describing how a call was ultimately resolved or left.

Standardizes call outcome for consistent reporting and analytics across agents and client companies.

Column	Data Type	Null	Default	Key
disposition_id	INT (Identity)	No	Auto-generated	PK
disposition_name	VARCHAR(50)	No	None	-
description	VARCHAR(300)	Yes	Not Stated	-

Primary Key	disposition_id
Foreign Keys	None
Unique Constraints	disposition_name
Indexes	IX_Dispositions_Name
Relationships	1:N to CallRecords.
Business Rules	A disposition cannot be removed while referenced by any CallRecords row.

Example: disposition_id: 2 | disposition_name: Resolved - First Contact

5.13 EscalationRules

Stores the configurable business conditions under which a call must be escalated.

Encodes the criteria evaluated during Business Rule Validation — for example high priority combined with negative sentiment — that trigger escalation tickets.

Column	Data Type	Null	Default	Key
escalation_rule_id	INT (Identity)	No	Auto-generated	PK

Column	Data Type	Null	Default	Key
rule_name	VARCHAR(150)	No	None	-
trigger_condition	VARCHAR(300)	No	None	-
severity	VARCHAR(20)	No	Medium	-
active	BOOLEAN	No	True	-

Primary Key	escalation_rule_id
Foreign Keys	None
Unique Constraints	rule_name
Indexes	IX_EscalationRules_Active
Relationships	1:N to Tickets.
Business Rules	Only active rules are evaluated for new tickets; inactive rules are retained for audit.

Example: escalation_rule_id: 7 | rule_name: Negative Sentiment High Priority | severity: High | active: True

5.14 OperationalStatistics

Stores aggregated, pre-computed reporting metrics used for operational analytics dashboards.

A denormalized reporting table populated by scheduled jobs that summarize CallRecords, Tickets, and ReviewQueue activity into daily/monthly rollups.

Column	Data Type	Null	Default	Key
stat_id	INT (Identity)	No	Auto-generated	PK
stat_date	DATE	No	None	-
calls_processed	INT	No	0	-
auto_saved	INT	No	0	-
review_count	INT	No	0	-
escalated	INT	No	0	-
category_distribution (JSON)	VARCHAR(MAX)	Yes	Not Stated	-
sentiment_distribution (JSON)	VARCHAR(MAX)	Yes	Not Stated	-

Primary Key	stat_id
Foreign Keys	None (aggregated from CallRecords, Tickets, ReviewQueue by scheduled jobs)
Unique Constraints	stat_date (one rollup row per day)
Indexes	IX_OperationalStatistics_Date
Relationships	Logically aggregates many CallRecords rows per date; no enforced FK since the relationship is computed, not transactional.
Business Rules	Daily rollups are recomputed if any CallRecords row for that date changes approval_status after initial aggregation.

	monthly_statistics values are derived by summing daily_statistics rows, not recalculated independently.
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Example: stat_id: 900 | stat_date: 2026-07-01 | calls_processed: 1,204 | auto_saved: 812 | review_count: 392 | escalated: 46

6. Relationships

One-to-One

Calls to CallRecords: exactly one structured record is produced per completed call, and a CallRecords row cannot exist without its parent Calls row.

One-to-Many

- ClientCompanies to Customers, Agents, and HandbookRules.
- Customers to Calls.
- Agents to Calls (as handling agent) and to ReviewQueue (as reviewer).
- CallRecords to Tickets and to ReviewQueue.
- Categories to Subcategories.

Many-to-One

From the child's perspective, CallRecords to Categories, Subcategories, Sentiments, and Dispositions are each many-to-one: many call records may share the same classification value.

Cascade, Delete & Update Behavior

- Delete behavior is restrictive by default — a parent row cannot be deleted while child rows reference it, preventing orphans.
- Soft deactivation (a status/active flag) is used instead of physical deletion for ClientCompanies, Agents, HandbookRules, and EscalationRules.
- Update behavior cascades on primary key changes only in rare key-migration cases; surrogate keys are otherwise immutable.
- CallRecords, Tickets, and ReviewQueue rows are never deleted — correction happens through the review workflow, which appends audit history.

Relationship Cardinality

Parent	Child	Cardinality
ClientCompanies	Customers	1:N
ClientCompanies	Agents	1:N
ClientCompanies	HandbookRules	1:N
Customers	Calls	1:N
Agents	Calls	1:N
Calls	CallRecords	1:1
Categories	Subcategories	1:N
Categories	CallRecords	1:N
Subcategories	CallRecords	1:N
Sentiments	CallRecords	1:N
Dispositions	CallRecords	1:N
HandbookRules	CallRecords	N:N (logical)
CallRecords	Tickets	1:N
CallRecords	ReviewQueue	1:N

Parent	Child	Cardinality
Agents	ReviewQueue	1:N
EscalationRules	Tickets	1:N

7. Constraints & Indexing

Primary & Foreign Keys

Every table uses a single-column surrogate integer primary key (identity/auto-increment), providing a stable, non-business-meaningful identifier. Foreign keys enforce every parent-child relationship in Section 6, ensuring CallRecords, Tickets, and ReviewQueue rows can never reference non-existent parents.

Unique Constraints & NOT NULL

Unique constraints prevent duplicate business identities, such as one CallRecords row per Calls row, one HandbookRules version per rule code, and one OperationalStatistics row per calendar date. Fields required for the platform to function (call_id, agent_id, priority, review_status, approval_status) are NOT NULL. Fields that may be genuinely unknown (customer_id, root_cause) are nullable and default to the literal value Not Stated rather than remaining empty.

CHECK Constraints

- priority must be one of Low, Normal, High, Critical.
- review_status must be one of Pending, In Review, Reviewed.
- approval_status must be one of Pending, Approved, Rejected.
- sentiment_score, when present, must be between -1.00 and 1.00.

Index Recommendations

Use Case	Recommended Index
Customer Lookup	IX_Customers_Phone, IX_Customers_Account
Review Queue	IX_ReviewQueue_Status, IX_ReviewQueue_Reviewer
Analytics	IX_OperationalStatistics_Date, IX_CallRecords_CreatedAt
Ticket Search	IX_Tickets_Status, IX_Tickets_Priority
Transcript Search	IX_Calls_TranscriptRef
Category Filtering	IX_CallRecords_Category (category_id, subcategory_id)

8. Normalization & Data Integrity

Normalization (3NF)

Every non-key attribute in each table depends on the whole primary key and nothing but the primary key. Reference values such as category names, sentiment labels, and dispositions are extracted into their own tables rather than repeated as free text on CallRecords.

Intentional Denormalization

OperationalStatistics is intentionally denormalized. It stores pre-aggregated counts and JSON distribution summaries that could, in principle, be derived on demand — this protects transactional performance by moving expensive aggregation off the live call-processing path onto a scheduled batch process.

Data Integrity Rules

No Orphan Records

Foreign keys and restrictive delete behavior ensure Tickets, ReviewQueue, and CallRecords rows can never exist without a valid parent Calls row.

Required Relationships

Every CallRecords row must link to a Calls row and an Agents row; every ReviewQueue row must link to a CallRecords row and a reviewing Agents row.

Mandatory Review

A CallRecords row cannot reach Approved unless review_status is Reviewed and a corresponding Approved ReviewQueue entry exists.

Mandatory Audit Trail

created_at, updated_at, reviewer_id, and edited_fields are populated for every reviewed record.

No Fabricated Data

AI extraction is prohibited from inferring facts not present in the transcript or a HandbookRules entry.

Unsupported Values Become Not Stated

Any field that cannot be confidently determined is stored as the literal text Not Stated rather than an empty string, a guess, or an ambiguous NULL.

9. Audit Trail & Security

Audit Trail

Original AI Draft

Initial values from the AI extraction pipeline are preserved in CallRecords at created_at, before any reviewer edits.

Human Edits

Any field changed during review is listed in ReviewQueue.edited_fields, showing the precise difference between the AI draft and the human-approved version.

Reviewer & Approval History

ReviewQueue.reviewer_id records who reviewed the record; status and approval_timestamp capture the outcome and timing of every decision, including rejections.

Timestamps

created_at / updated_at on CallRecords, plus review_start / review_end on ReviewQueue, provide a full timeline for each record.

Decision History

Tickets.decision_reason preserves the Decision Engine's rationale at the moment of routing, independent of any later human override.

Security Considerations

Role-Based Access Control

Access to Customers, CallRecords, and ReviewQueue is governed by role, so agents, reviewers, supervisors, and analytics users see only appropriate data.

Data Privacy

Customer identifying fields (full_name, phone_number, email, account_number) are isolated in Customers so sensitive personal data can be controlled and audited separately from operational data.

Least Privilege

Write access to reference tables (Categories, Dispositions, EscalationRules) should be restricted to platform administrators.

Encryption

Sensitive columns, particularly customer contact details, should be encrypted at rest, and connections should use encryption in transit.

Audit Logs

Database-level audit logging should complement the application-level audit trail, capturing direct data access outside the normal workflow.

10. Scalability & Future Enhancements

Scalability Considerations

Millions of Calls

Calls and CallRecords use narrow, indexed integer keys and are designed for partitioning by date or client company as volume grows.

Thousands of Customers

Customers is indexed on phone_number and account_number to keep Customer Lookup fast regardless of total customer count.

Growing Handbook

HandbookRules supports versioning, letting the knowledge base expand without invalidating historical validation decisions.

Large Analytics Datasets

OperationalStatistics absorbs analytics load through pre-aggregation, keeping dashboard queries fast even at hundreds of millions of transactional rows.

Future AI Models

Nullable, independently indexed classification columns allow new or improved AI models to populate the same structure without schema changes.

Future Enhancements

- Version history for CallRecords, preserving every reviewer edit as a discrete versioned snapshot.
- Multi-language support for transcripts, summaries, and handbook content.
- Multiple review levels, allowing supervisor sign-off for high-severity cases.
- Quality scoring of AI extraction accuracy, tracked per record and per model version.
- A feedback loop that routes reviewer corrections back into model evaluation datasets.
- Model performance tracking, correlating outcomes with the AI model version that produced the extraction.

11. Appendix

11.1 Entity Summary

Entity	Purpose
ClientCompanies	Stores the organizations (contact centers or brands) that subscribe to the After-Call Automation Platform.
Customers	Stores identifying and contact information for the end customers who place support calls.
Agents	Stores the support agents who handle calls and, where applicable, act as reviewers.
Calls	Stores metadata for each completed customer support call, including the transcript reference.
CallRecords	Stores the structured, AI-extracted, and human-reviewed documentation produced for each call — the core business entity of the platform.
Tickets	Stores the follow-up work item created when a call requires action beyond the call itself, including the Decision Engine's reasoning.
ReviewQueue	Stores every human review task, tracking who reviewed a record, what was changed, and when it was approved.
HandbookRules	Stores the documentation, policy, and handbook content used for retrieval during the Documentation Retrieval step.
Categories	Stores the top-level classification values used to categorize a call's subject matter.
Subcategories	Stores the second-level classification values that refine a Categories value.
Sentiments	Stores the controlled set of sentiment labels applied to a call by the AI extraction pipeline.
Dispositions	Stores the controlled set of outcome labels describing how a call was ultimately resolved or left.
EscalationRules	Stores the configurable business conditions under which a call must be escalated.
OperationalStatistics	Stores aggregated, pre-computed reporting metrics used for operational analytics dashboards.

11.2 Glossary

Term	Definition
Auto Save	A business decision indicating a record is eligible for final storage without further human review.
Route to Review	A business decision sending a record into the ReviewQueue for human validation.
Escalation	A business decision indicating a record requires elevated handling due to severity or policy sensitivity.
Not Stated	The controlled value stored when information cannot be confidently determined from the transcript or handbook.
Human-in-the-Loop	A workflow design requiring human review and approval before a system-generated output becomes final.

Term	Definition
Decision Engine	The business rule component that evaluates a record and produces a routing decision and decision_reason.

11.3 Abbreviations

Abbreviation	Meaning
ERD	Entity Relationship Design
PK	Primary Key
FK	Foreign Key
3NF	Third Normal Form
RBAC	Role-Based Access Control
AI	Artificial Intelligence